SkillNavora

MODULE 6 HANDBOOK

Module 6: SQL for Salesforce Marketing Cloud

6 sections, 62 topics explained, plus the detailed lessons, examples, exercises and official links.

Salesforce Marketing Cloud (SFMC) Academy. Verify product names, editions, licences and certifications on Salesforce's official pages.

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Edition: November 2026. Practice labs marked "account" need a provisioned Marketing Cloud account. Other labs use sample files or paper.

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Module 6: SQL for Salesforce Marketing Cloud

Level: Beginner to intermediate. SQL in SFMC: query activities, the supported syntax, aggregation, joins, data views and common audience queries.

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Detailed lessons

Module 6

6.1 SQL fundamentals in SFMC

WHAT IS IT?

> SFMC SQL runs in Query Activities inside Automation Studio. It uses a T-SQL style syntax with some restrictions compared with SQL Server. Core clauses: SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY.

WHY IS IT IMPORTANT?

> Audiences, reports and suppression lists are almost always built with SQL in SFMC.

HOW DOES IT WORK AND THE STEPS

1. Write the query in the Query Activity.
2. Set a target data extension and an update option (overwrite or append).
3. Run the activity, then check the row count in the target.

REAL-WORLD EXAMPLE

> Select active customers from the Customers data extension into an Audience data extension.

COMMON ERRORS

Writing a query that works in SQL Server but is not supported in SFMC. Always test in the Query Activity.

INTERVIEW QUESTION

What is a Query Activity?

HANDS-ON EXERCISE

Write a query that selects customers in one city and counts them.

OFFICIAL DOCUMENTATION AND FURTHER READING

- SFMC SQL reference (Salesforce docs)

6.2 Aggregation, joins and CTEs

WHAT IS IT?

> Aggregates (COUNT, SUM, AVG, MIN, MAX) with GROUP BY summarise data. JOIN combines data extensions and system data views. Common table expressions (WITH) name steps. Check the official SQL restrictions for the exact features your account supports.

WHY IS IT IMPORTANT?

> Joins and aggregates produce segments and reports; a wrong join silently changes audience size.

HOW DOES IT WORK AND THE STEPS

1. Use INNER JOIN when both sides must match.
2. Use LEFT JOIN to keep all customers and show missing orders as NULL.
3. Count distinct customers, not rows, when a customer has many orders.

REAL-WORLD EXAMPLE

> Customers with no order in 90 days: LEFT JOIN orders and keep rows where the order is NULL.

CODE EXAMPLE, WITH LINE-BY-LINE NOTES

```
-- Customers with no order in the last 90 days
SELECT c.SubscriberKey, c.EmailAddress
FROM Customers c
LEFT JOIN Orders o
  ON o.SubscriberKey = c.SubscriberKey
  AND o.OrderDate ⩾ DATEADD(day, -90, GETDATE())
WHERE o.OrderID IS NULL;
-- Line by line: LEFT JOIN keeps every customer; the date condition is inside the join so only
recent orders match;
-- WHERE o.OrderID IS NULL keeps customers with no matching recent order.
```

COMMON ERRORS

Counting rows where you need distinct customers.

INTERVIEW QUESTION

Explain the difference between INNER JOIN and LEFT JOIN.

HANDS-ON EXERCISE

Write the 90-day inactivity query and explain each line.

OFFICIAL DOCUMENTATION AND FURTHER READING

- SFMC SQL reference (Salesforce docs)

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6.3 SFMC data views

WHAT IS IT?

> System data views hold sends, opens, clicks, bounces, unsubscribes, jobs, journey activity and SMS tracking. Examples: _Subscribers, _Sent, _Open, _Click, _Bounce, _Unsubscribe, _Job, _Journey, _JourneyActivity, _SMSMessageTracking, _Complaint, _BusinessUnitUnsubscribes. Each has its own keys and retention limit.

WHY IS IT IMPORTANT?

> Engagement analysis and suppression depend on these views. Retention limits mean old data may be gone, so keep your own history.

HOW DOES IT WORK AND THE STEPS

1. Check the view's documentation for keys and retention.
2. Join on SubscriberKey and JobID where the view supports them.
3. Export long-term history to your own data extension.

REAL-WORLD EXAMPLE

> Last-open date per subscriber: SELECT SubscriberKey, MAX(EventDate) FROM _Open GROUP BY SubscriberKey.

COMMON ERRORS

Assuming a data view keeps data forever.

INTERVIEW QUESTION

What is _Open used for and what limits apply?

HANDS-ON EXERCISE

List the data views you would use for an engagement report and state each key and retention limit (verify on the docs).

OFFICIAL DOCUMENTATION AND FURTHER READING

- SFMC data views (Salesforce docs)

6.1 SQL Fundamentals: every topic explained

Introduction to SQL. SQL selects and shapes data. In SFMC it runs in Query Activities and writes results to a target data extension.

SELECT. Chooses the columns to return.

FROM. Names the data extension or data view to read.

WHERE. Filters rows by a condition.

DISTINCT. Removes duplicate rows from the result.

ORDER BY. Sorts the result. In SFMC Query Activities ordering is often not needed because results go to a table.

TOP. Returns only the first N rows (T-SQL style), often with ORDER BY.

AND and OR. Combine conditions; AND needs both, OR needs either.

IN. Matches any value in a list.

BETWEEN. Matches a range, inclusive at both ends.

LIKE. Pattern match with % (any characters) and _ (one character).

IS NULL. Finds missing values. NULL is not equal to anything, so use IS NULL.

CASE WHEN. If-then logic inside a query to create labels or segments.

6.2 SQL Aggregations: every topic explained

COUNT. Counts rows; COUNT(DISTINCT x) counts unique values.

SUM. Adds numeric values.

AVG. Average of numeric values.

MIN. Smallest value.

MAX. Largest value.

GROUP BY. Groups rows to aggregate per group.

HAVING. Filters groups after aggregation.

6.3 SQL Joins: every topic explained

INNER JOIN. Keeps rows that match in both tables.

LEFT JOIN. Keeps all rows from the left table, matched rows from the right or NULL.

RIGHT JOIN. Keeps all rows from the right table. Usually rewritten as a LEFT JOIN for clarity.

FULL OUTER JOIN concepts. Keeps all rows from both sides. Check support and performance before using it in SFMC.

SELF JOIN. Joins a table to itself, for example to compare rows of the same customer.

Multiple joins. Chains several joins; check row counts after each join.

Joining data extensions. Joins customers to orders or preferences on the subscriber key.

Joining system data views. Joins your data to data views such as _Open on SubscriberKey or JobID.

6.4 Advanced SQL: every topic explained

Subqueries. A query inside another query, used in WHERE or FROM.

Common Table Expressions. Named temporary results with WITH that make long queries readable. Check SFMC's SQL reference for support in your account.

Window functions. Functions that compute over related rows without collapsing them.

ROW_NUMBER. Numbers rows within a partition; used to keep the latest row per customer.

RANK. Ranks with gaps after ties.

DENSE_RANK. Ranks without gaps after ties.

PARTITION BY. Splits rows into groups for window functions.

LAG and LEAD concepts. Read the previous or next row's value. Check support before relying on them.

Date functions. GETDATE, DATEADD, DATEDIFF and CONVERT for date work.

String functions. LEFT, RIGHT, SUBSTRING, LEN, UPPER, LOWER, LTRIM and RTRIM.

Conversion functions. CAST and CONVERT change data types.

Null handling. ISNULL and COALESCE replace NULL values.

6.5 SFMC Data Views: every topic explained

_Subscribers. Subscriber records with status and dates. Check the docs for availability per business unit.

_Sent. One row per email sent, with JobID, SubscriberKey and EventDate.

_Open. One row per open, with event date and unique flag.

_Click. One row per click, with link URL and event date.

_Bounce. Bounce events with category and type (hard or soft).

_Unsubscribe. Unsubscribe events.

_Job. Send jobs with email name, subject and send dates.

_Journey. Journey versions and status.

_JourneyActivity. Activities inside journeys, linked to JourneyVersionID.

_SMSMessageTracking. SMS send and delivery tracking.

_Complaint. Spam complaints from mailbox providers.

_BusinessUnitUnsubscribes. Unsubscribes at business-unit level.

6.6 SFMC SQL Use Cases: every topic explained

Subscriber segmentation. Selecting audiences by attributes, behaviour and consent.

Campaign performance analysis. Sends, opens, clicks and bounces per job from data views.

Email engagement analysis. Open and click history per subscriber.

Bounce analysis. Bounce reasons by domain and campaign to fix list quality.

Unsubscribe analysis. Unsubscribe trends by campaign and segment.

Customer inactivity. Customers with no open, click or purchase in a set period.

Journey entry segmentation. Audiences built for a journey's data extension entry source.

Purchase history analysis. Orders by customer, product and date for targeting.

Customer retention. Repeat purchase and lapse measures.

Re-engagement audiences. Inactive but recoverable customers for win-back journeys.